The Nature of Vortex Ring State



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At low forward speed, when a helicopter's descent rate approaches the downward speed of its wake, the blade tip vortices organize in recirculating patterns beneath the rotor. A stable, ring-like wake structure emerges. Vortex ring state (VRS) is detected when the toroidal wake formation enters a buildup phase where organized vorticity accumulates more rapidly than it dissipates. The highly organized VRS wake dominates the descending rotor's inflow field, reducing blade angle of attack and creating thrust loss. The main event, observed in piloted rotorcraft during VRS, occurs when the ring makes a single transition through the rotor plane from below. Steady speed wind tunnel tests do not appropriately simulate VRS because thrust loss does not lead to higher descent rates in a wind tunnel. A vortex ring can only propel itself in a single direction. Thus, prior to transition, a curved flight trajectory, or maneuver, will tend to preclude or suppress VRS.

Dominance of the Vortex Wake

References 1 and 2 describe a multiphase effort that dedicated over 100 flight test hours to the piloted exploration of vortex ring state (VRS). From the onset, it was apparent that wind tunnel testing had not fully explained many of the observations found in flight test. Numerous discussions led to the development of a flight test perspective on the nature of VRS. On this basis, a model supporting test observations was developed and is put forward. At its heart, VRS is primarily concerned with the rotor blade tip vortices and how they influence the rotor in a low forward airspeed descent. VRS investigation, therefore, must focus on understanding the nature of rotor blade tip vortices.

While normally invisible, tip vortices are revealed on certain humid days, as seen for hovering and low-speed forward flight in Fig. 1. Even under ideal conditions, it is quite difficult to visualize tip vortices that are more than a few rotor revolutions old. The peak internal velocities subside as the vortex core ages. Eventually, the core's low internal pressure begins to recover, visible moisture returns to the flow, and the helical pattern disappears. This observed dissipation in the near wake should not be interpreted as a reduction in vortex strength. When a vortex core dissipates, practically all the circulation it contains remains in the flow, just distributed over a larger region. Thus, the vortex continues to influence the rotor in accordance with its distributed strength, orientation, and proximity to the blades.

The persistence of tip vortices near the rotor has always been the primary challenge to rotary wing flight and is a factor that fixed wing aerodynamicists may not fully appreciate. To illustrate, the idealized helical vortex wake cross-section for a hovering rotor, familiar to most rotary wing aerodynamicists, is examined in Fig. 2. The circulatory velocity of each vortex core extends its influence in a manner that drops off inversely with radial distance from the core center The cumulative effect of multiple, organized vortices on the surrounding flow is handled by the superposition principle, wherein the local induced flow velocity becomes the sum of the many individual vortex contributions.

The helical wake represents a set of continuous curved vortex filaments—one for each blade. In this respect, superposition is applied according to the Biot–Savart law, where the induced velocity at any point in the flow becomes the integrated contribution of each wake filament according to its strength, orientation, and location. Because the vortex filaments are organized in a well-defined structure, superposition works to multiply the effect of the vortices throughout the interior of the wake column and at the rotor plane. The combined influence of the vortices in the wake column is predicted by simple momentum theory, which shows that the induced velocity at the rotor is proportional to the square root of rotor thrust.

The influence of the tip vortices on the interior of the wake column continues to build downstream from the rotor. Ultimately, a flow speed that is twice the induced flow at the rotor disk is attained. The increase in speed occurs because the downstream flow is influenced by vortices that are both above and below the point of interest. Conversely, the rotor plane sees less inflow because there are no vortices above the rotor to contribute to the local flow. This same observation explains wake contraction. At the disk, a vortex is emitted at the tip of the blade. All filaments below the newest one produce a velocity that is directed inward as far as the newest tip vortex is concerned. By symmetry, the entire upper portion of the wake column is subject to a contracting velocity field. As the filaments move downward, their contraction ceases as more and more filaments reside above the point of interest to provide a "balancing" outward radial velocity component that cancels the wake contraction.

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Fig. 1. The key to understanding VRS is recognizing the preeminence of tip vortices.

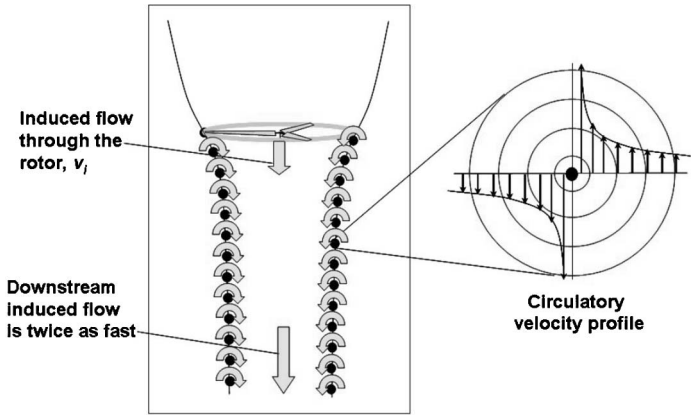


Fig. 2. Because the wake filaments are organized in a column, the circulation of individual elements work together, producing the net induced flow (v_i) at the rotor plane and twice v_i downstream.

The combined helical tip vortex filaments have a powerful effect on the flow. The influence can be put into perspective by considering that a segment of the helical tip vortex (of length equal to the rotor radius) influences the local flow to the same extent as the bound circulation of an entire rotor blade. Yet, while there are only a few blades on any given rotor, there are hundreds of equivalent "blade lengths" of tip vortex filaments comprising the helical wake. The perhaps striking conclusion is that rotor blades do not produce significant airflow. Rotors and fans produce tip vortices, and these vortices are responsible for the flow sensed downstream of the rotor disk. This is why there is so little wind on the upstream side of a fan.

This last observation is not intuitive from the principles of simple momentum theory and leads to the conclusion that the well-defined slipstream depicted for the hovering rotor cannot appear at all like the usual illustration of Fig. 2. The flow above the rotor is not bounded by a slipstream, as there is no mechanism for producing a defined shear layer where there are no vortices. Consequently, streamlines on the intake side of a hovering rotor approach the rotor disk from all directions, including below the rotor plane (as supported by experimental measurements in Ref. 3 and elsewhere). The true flow pattern, however, makes the derivation of simple momentum theory a challenge, hence, the "well-defined slipstream" extending far above the rotor disk.

The discrepancy between simple momentum theory's slipstream boundary and the actual flow near a rotor leads to more serious issues in the treatment of VRS. Simple momentum theory remains a surprisingly useful tool for assessing the hover, climb, and windmilling performance of an idealized rotor. Yet, its inability to describe flow boundaries hints at its primary weakness, which is that it cannot establish the paramount role of wake vorticity in governing the rotor performance. As Fig. 2 illustrates, every rotor designed for vertical flight must operate in a flow that is dominated by the vortex wake. This observation is the key for developing a rational understanding of VRS. The existence of the wake creates a significant hover power requirement that is captured by the inflow of the idealized rotor in simple momentum theory. This "nonnegotiable" penalty means that improvements in rotor blade airfoil design have a comparatively small effect on increasing the overall hover performance of rotors.

In a very general sense, wake vorticity never benefits the rotor, and when it is highly organized and concentrated, the performance penalty is significant. Figure 2 shows that the vortex-induced inflow requires the rotor blades to overcome a downwash field (v_i), just to maintain hovering flight. In this respect, the inflow is equivalent to an effective climb rate imposed on a rotor that is, in reality, just maintaining its vertical position. The downwash penalty is illustrated in Fig. 3, which is the familiar velocity diagram representing the rotor blade airfoil in hover. It is seen that the resultant velocity at any blade element is the vector sum of its rotational speed (Ωr) and the wake-induced velocity, v_i. The substantial result is that the hovering rotor blade airfoil is forced to operate at an effective angle of attack that is considerably lower than its geometric pitch angle. The point should be underscored, because the airfoil lift is proportional only to the effective angle of attack. This observation forms the basis of Prandtl's hypothesis that allows aerodynamicists to estimate the lift of a three-dimensional finite wing using two-dimensional airfoil data.

While lift is proportional to the effective angle of attack, the Kutta–Joukowski law specifies that the aerodynamic force vector (excluding

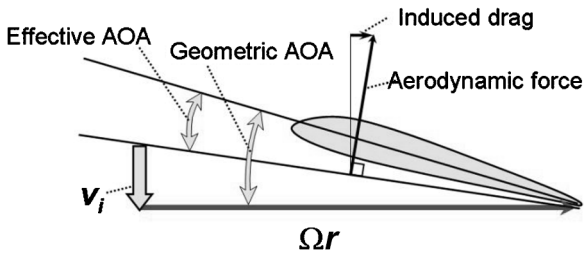


Fig. 3. Inflow forces the rotor blade airfoil to operate at reduced lift and with an induced drag penalty.

profile drag) must be normal to the local relative wind. In Fig. 3, the "backward tilt" of the aerodynamic force vector produces the well-known induced drag component that is the source of most of the helicopter's power requirement in hover and low-speed forward flight.

The Dynamics of Descent

To fully appreciate the rotor wake in a descent, it is important to make a few observations regarding the behavior of vortices that are in close proximity to one another. The classical depiction of the rotor wake as a smoothly contracting cylindrical column is a reference point for many readers. However, because of local interactions, the vortex wake does not actually retain a smooth trajectory for its helical filaments. Within a few revolutions, the vortices interact with each other, merging and corotating as discussed in Ref. 3. In a climb, the geometric distortions can be greatly diminished because the filaments are spaced further apart due to the higher inflow provided by a climb rate. The greater spacing means that the influence of vortices on each other is less, and therefore, any distortions will take more time to develop. Conversely, when a rotor starts to descend, the downward-traveling wake competes with the vertical velocity of the rotor, V_z (positive for climb, negative for descent). In the helicopter reference frame, the descent velocity would appear to reduce the net inflow ($v_i + V_z$). However, the vertical spacing between helical layers of the tip vortex filaments will be compressed. This compression not only increases the influence of the wake on the rotor (represented by the quantity v_i), but also causes the helical wake pattern to begin distorting closer to the rotor.

The initial wake distortion does not necessarily impact the rotor performance because the same amount of vorticity is still in the flow, and the influence on the rotor (in an integrated sense) is not substantially different from the idealized vortex wake geometry. However, the bundling of filaments has the effect of locally expanding the rotor wake. Later, in the modeling section of this paper, the wake is replaced by a series of simple vortex rings. It will be shown that in a descent, wake expansion promotes the reversal of the wake trajectory (toward the rotor) and the initiation of VRS.

For low descent rates, the wake pattern close to the rotor remains intact, and wake bundling occurs farther downstream, as illustrated in Fig. 4. When a descent is initiated, compression of the wake starts with

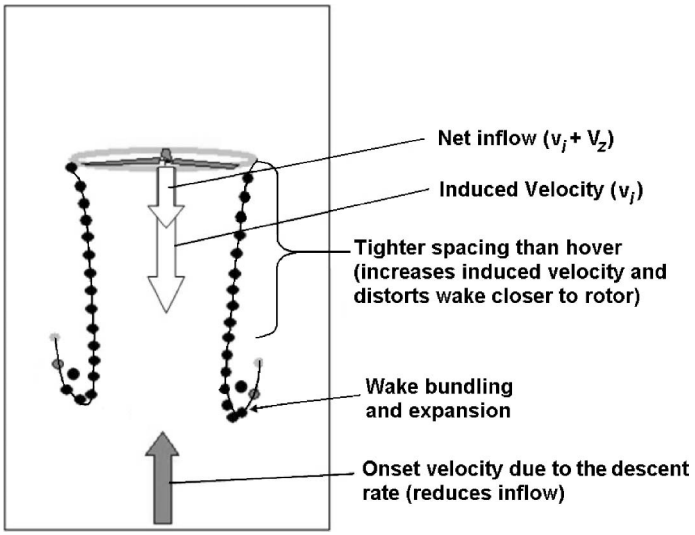


Fig. 4. The descent velocity competes with the wake induced-velocity to produce the net rotor inflow.

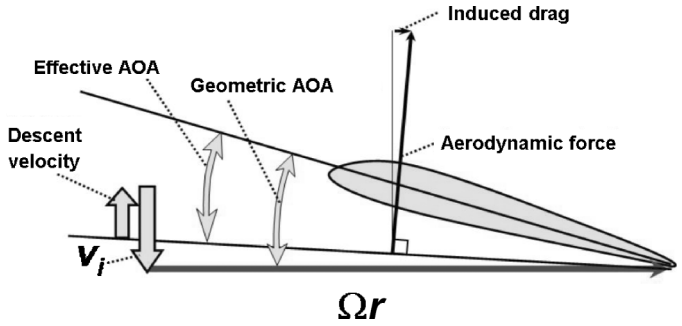


Fig. 5. At fixed collective pitch, the effect of low descent rate is to increase the effective angle of attack and decrease the induced drag.

the first few turns of the rotor and works its way through the lower wake column. In a low descent rate, rotor thrust departs only slightly from simple momentum theory. While the effect of wake compression tends to increase the rotor inflow (due to having more organized vortices closer to the rotor), the descent velocity more than compensates by directly opposing the rotor inflow. In terms of the effect on rotor thrust, reduced inflow is synonymous with an increase in blade effective angle of attack. Wind tunnel tests of rotors in steady vertical descent such as Castles and Gray (Ref. 4), or Betzina (Ref. 5), indicate that the initial stages of a fixed-collective pitch descent produce a slight reduction to the net rotor inflow (compared to the hover value), corresponding to an increase in rotor thrust.

Figure 5 develops the blade velocity diagram for this low descent rate. While the vortex-induced component of rotor inflow (v_i) has increased relative to hover, the direct effect of the opposing descent rate produces a beneficial net reduction in rotor inflow. Thus, for low descent rates produced in a wind tunnel at fixed collective blade pitch, the effective blade angle of attack increases and thrust goes up proportionately. In Johnson's work (Ref. 6), this stage of descent consists of a positive damping relationship between the descent rate and rotor thrust, similar to conditions in a hover or climb. For a free-flying rotorcraft, if a disturbance initiates an increased descent rate at this low descent rate, the rotor responds with an inherent increase in thrust that tends to arrest the increased descent rate without pilot intervention.

From an induced drag perspective, low descent rates reduce the "rearward" tilt of the aerodynamic force vector as shown in Fig. 5. In the case of a model rotor in a wind tunnel, initiating a low descent rate will produce additional thrust for a constant collective pitch, without requiring additional power. For the free-flying helicopter (operating at a fixed $1g$ thrust level), a reduction in rotor mast torque is the obvious effect of achieving a steady low descent rate.

As the descent rate continues to increase, additional compression takes place between successive layers of the helical wake. As before, this compression places more wake vorticity close to the rotor and tends to further increase inflow. However, the fundamental dynamic of VRS involves the tendency of closely spaced vortices to completely reshape the cylindrical wake and form a stable system of corotating/merging wake elements. The initiation of this behavior is depicted in Fig. 6, which shows the formation of recirculating vortices contained roughly within a toroidal structure beneath the descending rotor. Individual vortices located below the formation are forced to expand outward by the prevailing flow induced by the circulation of the torus.

Any expanded wake element is itself like a vortex ring with a larger diameter. Diameter growth reduces the ability of that vortex ring element to propel itself downward (conservation of vorticity). This effect will become clearer when the "ring emitter" wake model is discussed later,

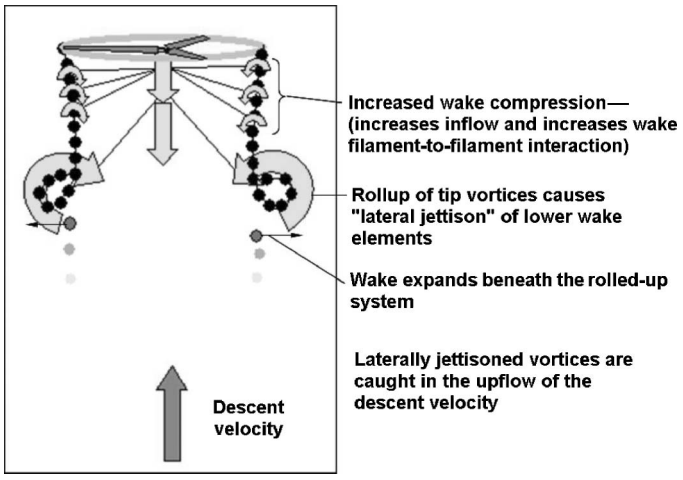


Fig. 6. The organized accumulation of tip vortices close to the rotor is the fundamental dynamic of VRS.

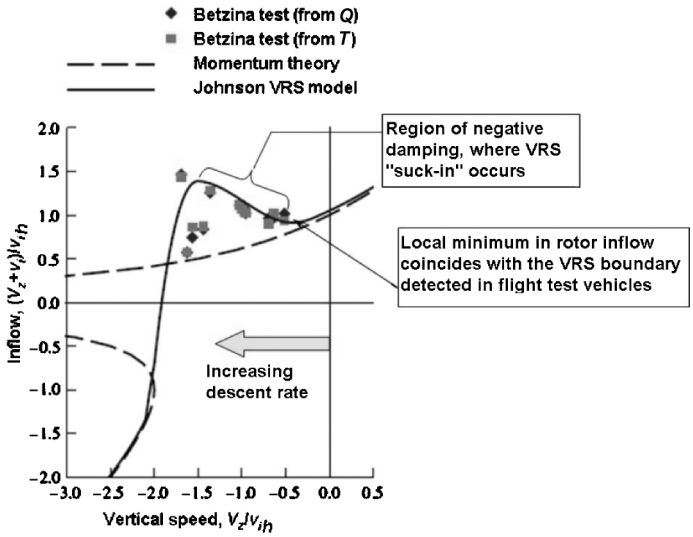


Fig. 7. In the context of VRS, inflow $(V_z + v_i)$ should be connected (in the reader's mind) to reduced blade angle of attack. Here, Johnson's work (from Ref. 6) shows that descent rate leads to negative damping with thrust in the VRS region.

and it is shown that "expanded" wake elements rise upward relative to the rotor in the free-stream flow. The key result of the toroidal ring formation is that the tip vortices have a mechanism for collecting in an organized pattern in close proximity to the rotor, where they compound the negative influence of the wake on rotor performance.

The notion of an organized, stable vortex ring structure produced from individual vortex filaments has a rigorous mathematical basis. Blackmore et al. (Ref. 7) examined the ability of multiple vortex rings to self-organize as a stable grouping, showing that a family of closely spaced, coaxial vortex rings will form a stable system of corotating ring elements. Although the physical rotor wake consists of helical vortex patterns, it is not difficult to imagine an equivalent simplified wake consisting purely of vortex rings, especially when descent rate flattens the helical pitch of the wake.

The formation of an accumulating vortex ring structure from individual vortex filaments marks a transition where the rotor no longer benefits from increased descent rate. In Johnson's work (Ref. 6), the rotor entering this stage of descent transitions into a regime where the relationship between thrust and descent rate is negatively damped. The relationship is best seen in a plot of rotor "total inflow" versus descent rate as shown in Fig. 7 from Ref. 6. To properly interpret Fig. 7, and the transition to VRS, it is critical to connect inflow (at the rotor disk) with a reduction in blade effective angle of attack. In Fig. 7, it is seen that when the vertical speed is near $V_z/v_{ih} = -0.6$, the slope reverses. In other words, higher descent rates (more negative than $V_z/v_{ih} = -0.6$) lead to more inflow, not less. Therefore, any increase in the descent rate (due to a disturbance) will further increase the rotor inflow, eroding the effective blade angle of attack, and rotor thrust. Based on flight test results reported by Taghizad et al. (Ref. 8) and the results of Refs. 1 and 2, the entrance into the "negative damping" region coincides with the descent rate where a free-flying rotorcraft will report entering VRS. Beyond this descent rate, the thrust-defeating effect of increased inflow feeds on itself and, in flight test, has been referred to as the VRS "suck-in." The term is apt because if the controls are held fixed, the rotor will increase its descent rate and pass directly through the vortex ring, emerging in a stable descent (with the ring above the rotor) as discussed in the next section.

In Fig. 7, Johnson applied certain assumptions to derive the "measured" inflow for Betzina's wind tunnel data. The result shows that the inflow for a wind tunnel test rotor will eventually exceed the hover inflow velocity by a factor of almost 1.5 when the descent rate reaches $V_z/v_{ih} = -1.5$. Since the inflow is composed of descent velocity that opposes the wake-induced contribution, it can be inferred from Fig. 7 that the influence (at the rotor) of the fully developed VRS wake is roughly three times greater than the induced velocity produced in a hover. In other words, if the net rotor inflow is $(V_z + v_i)/v_{ih} = 1.5$ with a descent rate of $V_z/v_{ih} = -1.5$, then the wake-induced contribution must be $v_i/v_{ih} = 3.0$. How is it that the rotor wake influence at the rotor can increase so dramatically in a descent? The answer lies in the formation of a stable, highly organized wake structure accumulating in close proximity to the rotor. This simple structure and its effect on the rotor is the essence of understanding VRS, and it will be shown that all of the primary features of VRS can be explained on this basis without resorting to speculative conjectures regarding wake distortions, or instabilities that do not relate to blade loading.

Simple Ring Emitter Model in Pure Vertical Flight

In the transition to VRS, simple momentum theory breaks down because the observed wake geometry violates the "well-defined slipstream" requirement for analyzing flow states. Nevertheless, because momentum theory molds our understanding of rotor behavior, misconceptions regarding VRS abound. These misconceptions include notions that there is no flow through the rotor disk in VRS or that blade stall is responsible for the thrust loss observed in VRS measurements. The effect of VRS on a rotor blade element (as supported by Fig. 7) is illustrated in Fig. 8. For a fixed collective pitch, developing VRS will reduce the blade's effective angle of attack because, in this regime, the induced velocity due to the accumulating organized vortex-ring wake grows faster than the opposing descent rate. As the net rotor inflow increases, the effective angle of attack of the blade decreases progressively. Rather than stalling, which would mean high angles of attack, the rotor in VRS loses thrust because of a reduction in the blade's effective angle of attack that comes with the high inflow produced by VRS. The fact that inflow increases is a testament to the high level of wake organization during VRS and should serve to counter any claims arguing that wake disorganization is a contributor to thrust loss in VRS. As the organized wake column in hover produces the hover-induced velocity, the highly organized vortex ring wake produces the inflow of VRS.

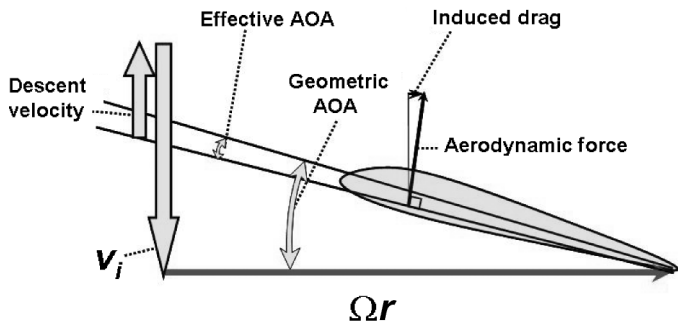


Fig. 8. As VRS develops, the influence of organized vorticity accumulation negates the benefit of descent rate and leads to a reduction in effective blade angle of attack.

Reducing VRS to its simplest component requires a model that recognizes the primacy of vorticity in influencing the rotor. The simplest model that can do this is a "vortex ring emitter." This model represents vertical flight by replacing the helical rotor wake with a sequence of individual vortex rings that are continually emitted, one for each blade, for each rotor revolution. The predictive capability of the model is fundamentally linked to its ability to not only capture sufficient revolutions of wake vorticity (up to $160/n_b$, where n_b is the number of blades) but to allow the resulting wake elements to self-organize. Models that fail to retain sufficient wake vorticity, or that do not allow wake self-organization to take place, or that do not recognize the time-variant (vorticity accumulating) phase of a descent, will not be able to examine the full spectrum of VRS. Before developing the model, it is important to review a few fundamentals of vortex ring behavior.

Vortex rings influence not only the surrounding flow but also themselves. As long as axial symmetry applies (as in pure vertical flight), each emitted ring will remain centered about the rotor's vertical axis. Each vortex ring, by virtue of its own self-influence, will propel itself along the axial direction (usually downward and away from the rotor). Each ring is influenced by all the other rings that comprise the wake. Individual rings will contract or expand in diameter as they move relative to the other wake elements. The basic behavior can be observed in an experiment that is probably familiar to both chain smokers and aerodynamicists. It involves puffing two smoke rings in rapid succession, one behind the other. While the two smoke rings propel themselves away from the smoker, they also interact with each other. The trailing ring tends to be contracted (in its diameter) by the circulatory flow from the leading vortex ring. Conversely, the leading vortex ring experiences an outward flow due to the trailing ring's influence. The vorticity of each ring is conserved; therefore, a diameter reduction will increase the self-induced velocity of the trailing vortex, while the self-induced velocity of the leading vortex ring is reduced. The experiment shows that the trailing vortex will overtake the leading vortex, passing completely within its circle. After this, the pair, continuing to be propelled in the axial direction, repeats the leapfrogging process of swapping the leading and trailing position.

In the simple wake model, emitted vortex rings become part of a Biot–Savart calculation that tracks the influence of each ring on itself and on the surrounding community of other rings. Axial symmetry is retained throughout the calculation, limiting the analysis to pure vertical flight (hover, climb, descent). Details governing the analysis of emitted vortex rings are provided in the Appendix. In the ring emitter model, rotor thrust is calculated using a simply articulated 11-segment blade element model that accounts for blade twist, collective, and the net inflow at each blade element to determine its effective angle of attack. Small angle assumptions are expressly avoided. Blade flapping response is permitted, but in axially symmetric conditions its inclusion does not materially impact the solution. In axisymmetric conditions, flapping cannot alter the symmetry of the wake or change the fluid mechanism that allows vortex rings to accumulate and organize. For this model, emitted rings contain 100% of the blade-bound circulation, and there is no distributed sheet of vorticity. For simplicity, the emitted vortex ring strength is based on the average blade lift determined from the net rotor thrust. The core diameter was kept small at 0.075% of the rotor radius. To simulate free-flight conditions, the model trims to a hover condition and then allows the user to adjust the collective pitch. Any change in thrust from the initial trim condition will cause a climb (or a descent rate) to develop. A wind tunnel mode feature is also available for correlation with the body of existing VRS data. Wind tunnels are, however, fundamentally different from a free-flying rotor where VRS is concerned. This is because the wind tunnel flow (responsible for descent rate) does not respond to the rotor thrust. For a fixed collective pitch, wind tunnel models will experience large thrust loss (relative to hover trim thrust) as VRS is approached. Conversely, a free-flying rotorcraft will fall faster if thrust is reduced. This changes the rotor wake buildup by adding a free-stream component to the dynamics. Because of this interaction, most of the free-flying rotor's descent will occur at close to $1g$ load factor conditions. Thrust loss, mitigated by downward acceleration, is how VRS is perceived in piloted rotorcraft. This free-body interaction does not take place in conventional wind tunnel experiments on VRS.

Using the ring emitter model, the free-flight vertical descent process is taken to fully developed VRS in the sequence identified in Fig. 9. At the top of the figure, the wake patterns for the emitted rings are shown as snapshots at key phases of the descent. The time history data show the effect of the sequential reductions in rotor collective pitch of Fig. 9(a). From a stable hover at time $= 0$, a 0.25-deg collective drop at $T = 25$ s initiates a small descent rate. The initial thrust loss is seen in Fig. 9(b). However, at low descent rates, the thrust loss is fully and automatically recovered as a steady descent rate develops (black curve in Fig. 9(a)). Over the next four collective pitch drops to $T = 325$ or $\theta = 13^\circ$, the main change in the observed wake pattern is that the tip vortex trajectory near the rotor tip (see corresponding insets in Fig. 9) transitions from an initial downward trajectory to an (initial) upward trajectory due to the free-stream influence.

This lingering of the tip vortices above the rotor plane steadily increases the wake influence on the rotor as seen by the rise in v_i/v_{ih} in Fig. 9(c) (recall that v_i represents wake influence at the rotor disk). Nevertheless, up to this stage of descent, the net inflow through the rotor ($V_z + v_i$) always returns to unity due to the beneficial effect (upflow) of the newly developed descent rate (see Fig. 9(c)). This is consistent with the positively damped region of Fig. 7 and indicates the full and complete restoration of $1g$ thrust in the rotor. A secondary effect of the new initial upward tip vortex trajectory is that it introduces a "rumble" into the rotor thrust (Fig. 9(b)). The "rumble" is due to the new tip vortices passing through the rotor plane (from above) as they propel themselves downward into the wake column. The resulting blade–vortex interactions are undoubtedly the cause of the unique acoustic signature reported by pilots who have approached the VRS condition at low forward speed.

Two additional collective drops (at $T = 400$ and 475 s) further increase the descent rate to $V_z/v_{ih} = -0.7$. At this time, the pattern of ring vortices (corresponding to a collective pitch of $\theta = 12.3^\circ$) is a considerably thicker column than previous patterns. Groupings of ring vortices that had appeared further down the wake (in hover and low-speed descent) now form closer to the rotor. As shown in Fig. 6, these groupings lead to the expansion of "lower" rings. As discussed in the smoke ring analogy, expansion of a vortex ring causes the ring's self-propelling (downward) speed to diminish, allowing the free-stream and inboard vortices to convect the expanded rings upward toward the rotor

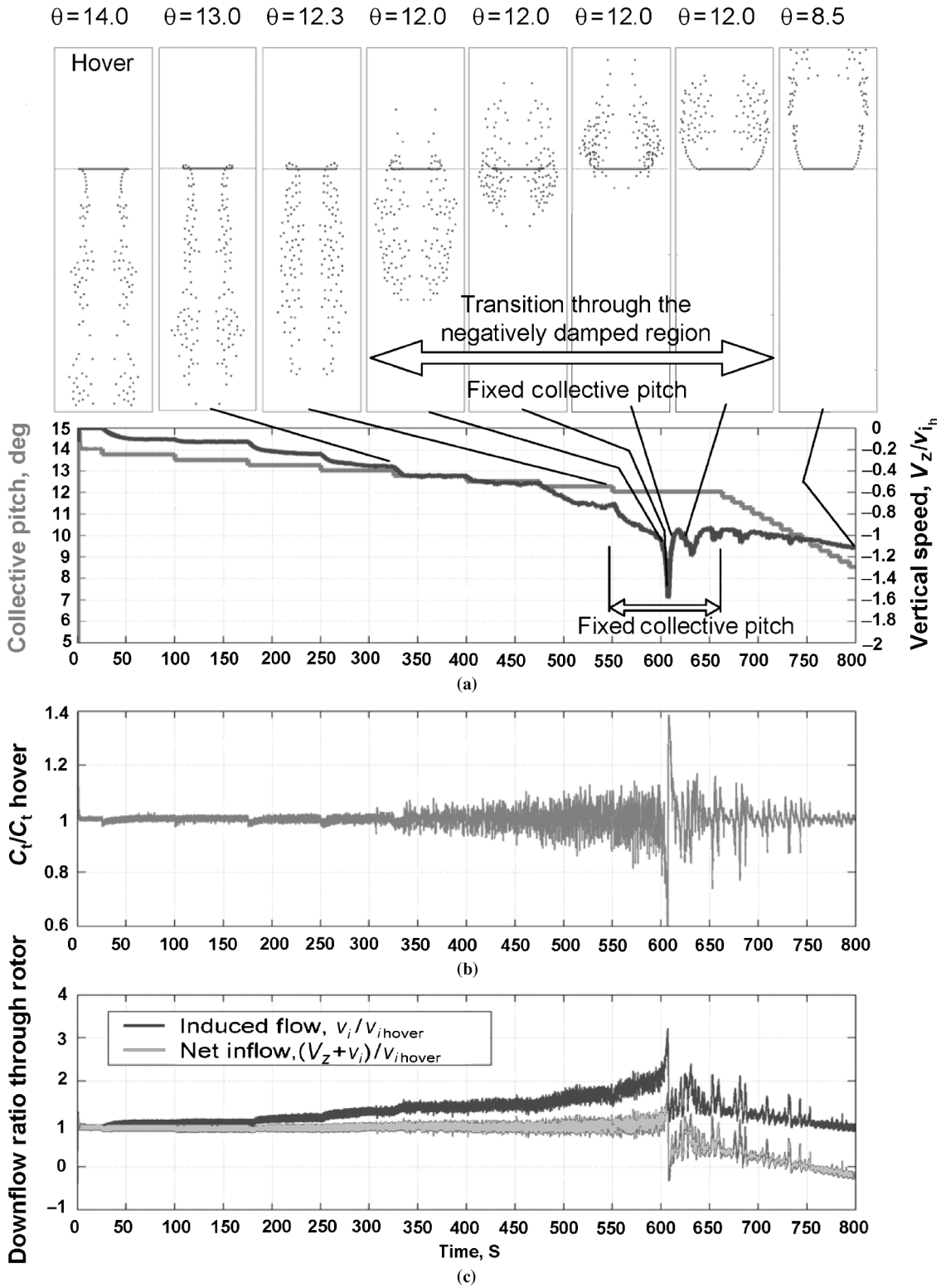


Fig. 9. In a free descent, VRS emerges after successive drops in collective pitch.

where they can again (negatively) influence rotor performance. The closer these groupings are to the rotor, the greater their propensity to increase the lower wake diameter and send vortices outward and upward toward the rotor, largely disrupting the cylindrical wake structure. When old tip vortices emerge above the rotor plane, they become entrained by the inflow pattern of the existing vortex wake system, are forced to contract, and regain their downward self-induced speed. Additional thrust rumble takes place as these vortices are reingested by the blades, as seen by the steadily increasing thrust "noise" in Fig. 9(b). It is important to note that for the collective pitch reductions prior to $T = 550$ s, stable conditions are reestablished. The initial accumulation of wake vorticity reaches a new equilibrium where organized vorticity convects away from

the rotor as rapidly as new vorticity is added by the rotor blades. The equilibrium of net wake vorticity allows inflow to stabilize and thrust to recover to $1g$ at the new, higher descent rate, consistent with positive thrust damping.

At $T = 550$ s, the collective drop (to $\theta = 12^\circ$) initiates the most significant phase of VRS (i.e., the phase with negative thrust damping with respect to descent rate). All prior collective pitch drops settled to a higher, but constant, descent rate. Now, with no further collective pitch reductions, the rotor marches toward fully developed VRS and four wake pattern pictures are needed to show the dynamic transition that takes place in comparatively few seconds. In essence, the wake pattern shifts from an open loop (where vortices escape the vicinity of the rotor) to a closed loop, where wake vortices accumulate in an organized recirculating pattern. The entire lower portion of the wake column ejects vortices outward and upward, returning them to the rotor. The closed return path developing at $\theta = 12^\circ$ represents the primary dynamic of VRS buildup, marked by a rapid growth in rotor inflow that is caused by a growing mass of highly organized vorticity. While each blade passage adds another vortex to the rapidly developing ring, the rotor is placed in a losing battle where inflow robs the blades of angle of attack and, therefore, lift. Once the rotor approaches the negatively damped region of Fig. 7, the thrust loss is so rapid that the rotor essentially falls through the ring, emerging with the ring above the rotor in a new, stable (positively damped) flow state (yet still under the influence of the vortex ring formation).

As the entire wake structure collapses into a toroidal formation, the rotor inflow rises sharply as seen in Fig. 9(c) near $T = 606$ s. To appreciate the impact of this developing vortex ring, it is worthwhile to consider that the hovering rotor distributed its wake vorticity over an extensive helical column (Fig. 2) to generate the rotor's hover-induced velocity field. In VRS, these same filaments are no longer distributed far below the rotor but are collapsed into a geometrically stable, highly organized ring structure in close proximity to the rotor. This highly ordered state multiplies the ability of the vortex filaments to produce inflow well beyond the hover value achieved by the distributed wake column. This is the reason that the inflow inferred from wind tunnel test data in Fig. 7 shows a 50% increase in net inflow compared to the hover value and surmises a tripling of the rotor wake strength (induced velocity) for fully developed VRS.

The presence of a high inflow state (caused by the accumulating vortex ring) forces the rotor blades to low angles of attack and thrust loss is the integrated effect. This transition phase of VRS correlates with the stability boundary identified by Johnson (Ref. 6). It is also related closely to the Newman boundary defined using the wake transport velocity criteria in Ref. 9. In experimental flight testing, the entry into the negative damping phase corresponds to the descent rate where the normal load factor suddenly drops and is similarly related to the roll-off boundary identified in tiltrotors. On the basis of Fig. 9, the negatively damped region of Fig. 7 can never be considered a flight regime where steady conditions could be achieved because the ring and rotor are exchanging positions.

The relationship of thrust to descent rate is negatively damped because in this phase, the ring is not only in an accumulating condition that increases inflow but is also in relative motion toward and through the rotor plane, where its proximity has a greater influence on the rotor. The closer the rotor approaches the ring, the more the rotor's inflow increases, the more it loses thrust, and the more rapidly it descends toward (and through) the ring. This portion of VRS produces the "fall through" sensation reported by pilots and is in line with the observation (in Fig. 9 at $T = 606$ s) that thrust (transiently) drops to 60% of $1g$ before ending in a stable flight condition with the ring above the rotor.

The transition (from a vortex ring below the rotor to a ring above the rotor) should not be confused with the transition between the climbing and windmilling branches of the momentum theory inflow curves. Figure 10 shows the "fixed collective" transition of the descent in Fig. 9 as a gray trace, with the other portions of the descent in black. It is clear that the ring transition in Fig. 9 takes place completely on the upper branch of the inflow plot. This is so because the transition involves the accumulation of many vortex filaments in close proximity to the rotor, with each element contributing to a high inflow state. As long as the inflow remains high (due to wake influence), the solution will remain on the upper branch. Even after passing through the rotor plane, the vortex accumulation remains close to, albeit above the rotor, causing the inflow state to remain high. However, with the ring formation above the rotor, and no longer accumulating, the relationship between descent rate and inflow/thrust (Fig. 7) is no longer negatively damped. As Fig. 9 shows, many more successive drops in collective pitch are needed to drive the rotor away from the vortex accumulation and onto the lower windmilling branch. This result is indicative of high stability, as seen from the steep "positive slope" that exists in Fig. 7 when transitioning between upper and lower branches of the rotor inflow/descent rate relationship.

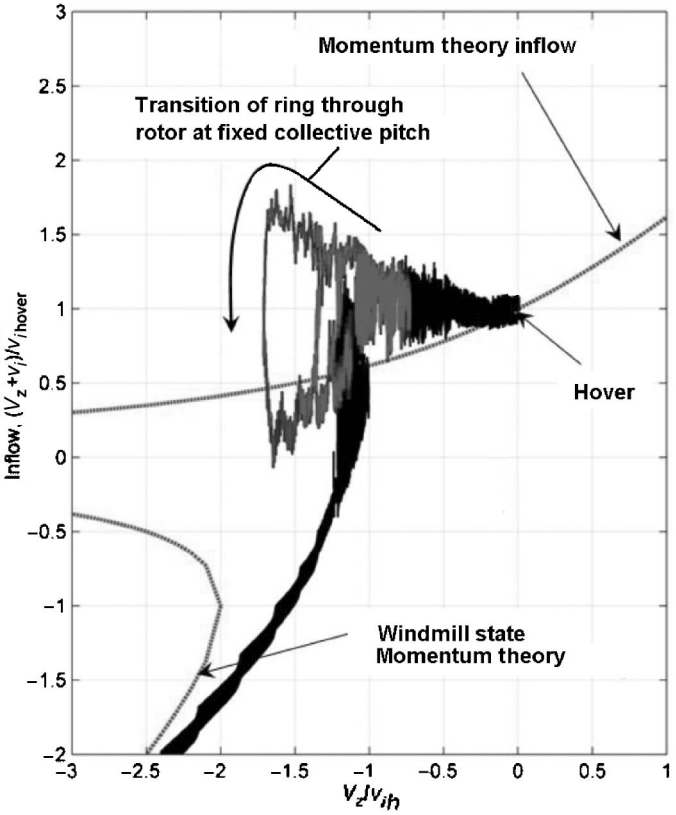


Fig. 10. The transition to simple momentum theory's lower branch (windmill state) is not related to the ring transitioning from below the rotor to above the rotor. A much higher descent rate (lower collective) is needed to clear the "ring formations" from above the rotor.

The escape (or accumulation) of vorticity from the rotor is the determining factor of VRS and whether or not the thrust is negatively damped with respect to descent rate. Two primary mechanisms allow vorticity to escape the rotor. The most obvious is the simple convection of the ring (wake element) away from the rotor. This is the mechanism seen in hover, climb, or forward flight. A second mechanism, pointed out by Ahlin and Brown (Ref. 10), also exists for eliminating vorticity from the inflow calculation. Essentially, organized vorticity (which contributes to the rotor

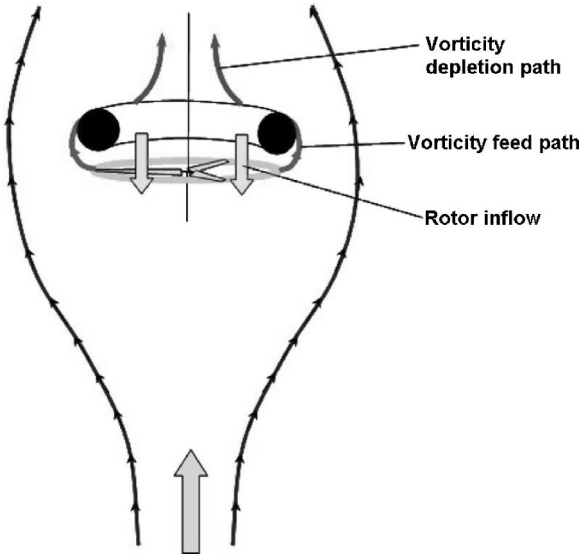


Fig. 11. Conceptual model of the last stage of VRS: An equivalent vortex ring with vorticity feed path, depletion path, and rotor inflow.

inflow) is reduced when free wake filaments distort from the horizontal plane and begin to contribute to swirl velocities rather than inflow. This mechanism for the depletion of organized vorticity could certainly play a role in limiting the ultimate strength in an accumulated vortex ring. It is probably evidence for arguing that VRS could be "defeated" by sufficient power and collective pitch range. Nevertheless, the fundamental character of VRS is not the so-called "destruction" of vorticity, but rather, it is the growth of organized vorticity and the associated inflow that causes thrust loss. In the negatively damped region, organized vorticity must accumulate faster than it is depleted, convected, dispersed, or destroyed; otherwise a positive thrust damping relationship would exist with descent rate.

In free flight, the "posttransition" vortex ring condition settles into a steady-state formation that can be conceptualized by the illustration of Fig. 11. Here, an equivalent ring of distributed vorticity resides above the rotor after transition. As with all free vortices, this vortex ring is not attached to the rotor system by any means. To maintain its position relative to the rotor, the vortex ring must propel itself downward through the air mass by virtue of its own self-induced velocity. In the same way that a smoke ring moves through still air, the vortex ring propels itself, chasing the rotor down hill and causing it to operate at an elevated sink rate, albeit at the full $1g$ thrust conditions. Lamb, Helmholtz, and numerous others have studied the self-propulsive speed of large core vortex rings. Relationships governing the self-induced speed are based on circulation strength, core diameter, and ring radius, as reported in the work of Saffman (Ref. 11) or Widnall et al. (Ref. 12). This self-propulsive speed is calculated numerically for each ring in the emitter model as given in the Appendix and is seen to agree with the results of Saffman (Ref. 11).

Once the ring formation has transitioned above the rotor, high stability (thrust damping with descent rate) is achieved. Thrust damping (with descent rate) also exists even if the vortex ring grows in strength (accumulates vorticity) transiently. An increase in strength will cause the ring to propel itself downward toward the rotor. The accumulated ring would overtake the rotor, except that the same ring causes the rotor to experience higher inflow. As the ring approaches the rotor, its influence on the rotor increases, producing the attendant reduction in blade effective angle of attack and thrust loss. With any thrust loss, the rotor accelerates downward, increasing the gap between the ring and the rotor and allowing thrust to recover. In this way, a stable feedback mechanism exists to maintain the relative position of the ring and rotor after transition occurs. This discussion (relating to Fig. 11) is in contrast to steady descent wind tunnel test data such as Ref. 13, and more recently by Stack et al. (Ref. 14). It will be shown that the thrust-modulating feedback mechanism (present for free flight) does not exist in fixed-speed wind tunnel tests, and therefore the effects of VRS, in terms of thrust changes and the unsteady flow patterns, will tend to be overstated in such tests.

The unsteady nature of VRS is often cited in background literature. More than 50 years ago, the flow visualization experiments of Drees and Hendal (Ref. 13) set the general impression of VRS as a rotor ingesting, scattering, and reingesting a chaotic wake formation. Yet, this impression is misleading where it concerns the relation to flight test data of rotorcraft. VRS requires the formation of a stable, highly organized vortex structure composed of accumulating wake elements. If this were not the case, a high inflow could not be achieved. As discussed next, the primary unsteadiness perceived in wind tunnel and water tunnel experiments of VRS is caused by a stable structure (the vortex ring) undergoing significant and repeated relative motion with respect to the rotor. In such tests, the chaotic impression is a fallout from testing the rotor at fixed descent rates that cannot represent a free-body rotorcraft. While the (fixed) tunnel speed sets the rotor's descent rate, the ring's descent rate (and its position relative to the rotor) remains ungoverned because the feedback mechanism controlling ring–rotor separation (in a free descent) is eliminated.

When the descent rate is held fixed in the negatively damped region, the vortex ring will accumulate vorticity to the point that its self-induced propulsive speed exceeds the rotor's (prescribed) descent rate. The ring propels itself downward (through and away from the rotor). The rotor inflow (caused by the ring) is negated, thrust recovers, and the free stream establishes a new wake pattern above the rotor. Quickly, a fresh vortex ring pattern emerges above the rotor, inflow builds, thrust is reduced, and the accumulating ring begins another cycle of passing downward through the rotor plane. Each time the ring approaches the rotor, thrust is reduced, but the rotor cannot accelerate (fall) away. Even though it experiences thrust loss, its "descent speed" is held fixed by the wind tunnel operation. The vortex ring will be able to propel itself downward directly through the rotor plane, dispersing vorticity below and repeating the process, as seen in flow visualization tests. With each cycle, a fresh vortex ring regenerates above the rotor and a new surge of inflow and thrust (loss) is created in a scenario that does not accurately represent the dynamics of free flight. Sustained thrust fluctuations on the order of the mean thrust level have been reported in such (fixed descent rate) tests.

To examine the wind tunnel mode experiment, the vortex ring emitter model is started in a windmilling state (high descent rate) with collective pitch held constant. Figure 12 shows an initial vertical descent rate of approximately two times the hover-induced velocity followed by sequential reductions in the (wind tunnel) descent speed. With the reductions in descent rate, the rotor thrust decreases because the blades see a reduction in effective angle of attack due to the reduced (upward) free-stream flow. As the descent rate is further reduced between time $=$ 30 s and time $=$ 70 s, a vortex ring begins to gather above the rotor. Because of the fixed number of rings used in the emitter model (80), the accumulated vortex ring eventually cannot become any stronger (old rings disappear from the calculation of inflow as rapidly as new rings are created). For this descent rate, the accumulated ring remains in a fixed position above the rotor. Reductions in descent rate can be continued, allowing the organizing ring to drive downward through the rotor plane. Alternatively, realizing that the 80-ring limitation (used in the emitter model) is arbitrary, more rings can be permitted in the calculation. In this case, the number of emitted

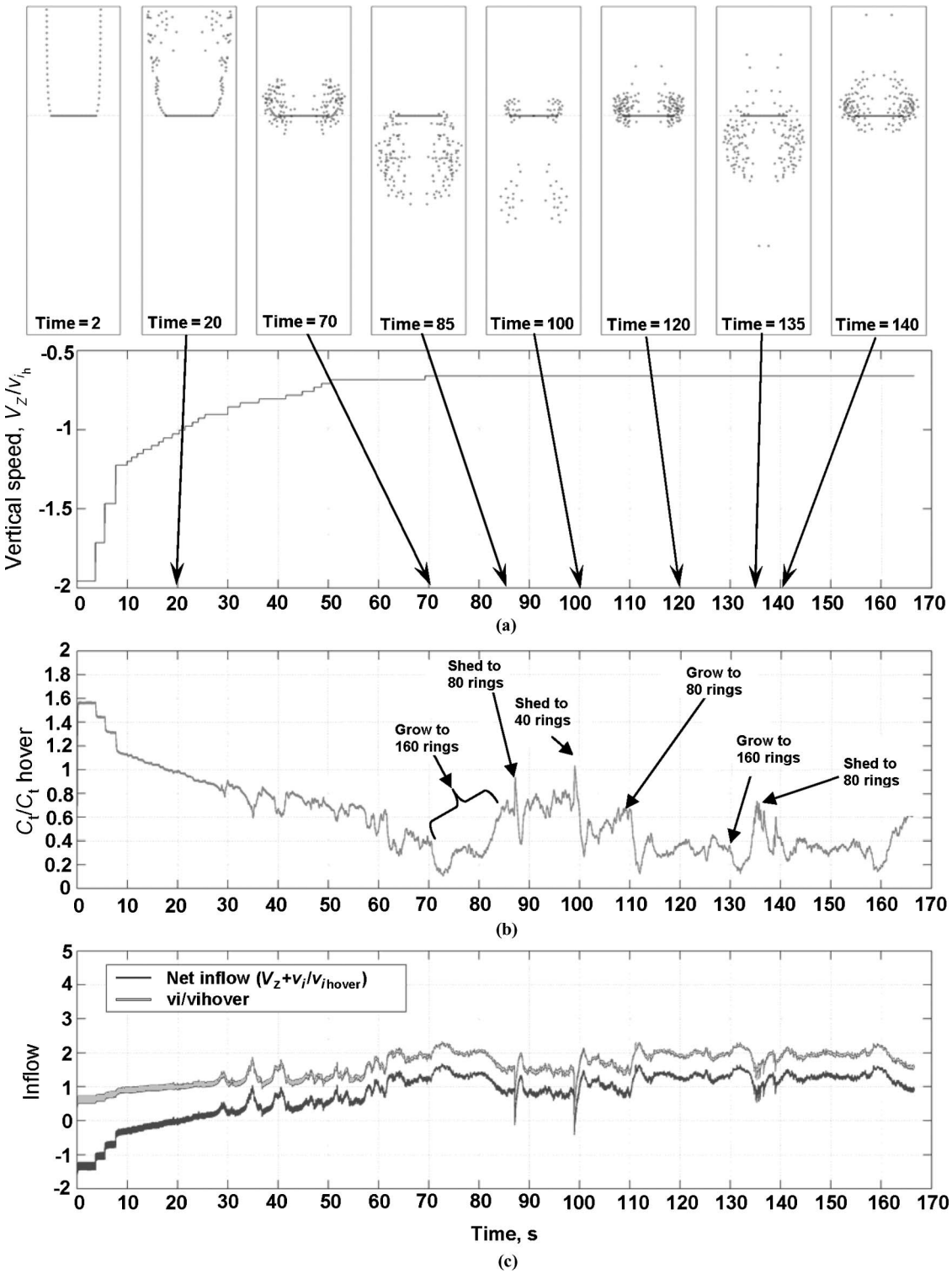


Fig. 12. By manually changing the number of accumulated rings (to simulate vorticity shedding), this wind tunnel mode simulation shows mean thrust loss, and repeated thrust surges as the ring continually reforms and passes through the rotor plane—overstating the effects of VRS compared to free descent.

rings is allowed to double to 160 starting at time = 70 s. The additional vorticity self-accumulates, having the immediate effect of charging the ring with sufficient vorticity that it is (by $T = 85$ s) propelled downward to a stable position below the rotor. Although there have been no changes to the descent rate since time = 70 s (tunnel speed is held fixed), there are significant thrust changes due purely to the accumulating ring and its relative motion through the rotor.

Although the fixed number of rings in the ring emitter model limits the strength of the vortex system, the mathematical stability (due to the forced axial symmetry of these calculations) will omit secondary effects

such as local wind and buffeting that would lead to wake disruption and the development of large asymmetries in the ring and surrounding flowfield. The model indicates that the vortex ring will accumulate all 160 emitted rings and stabilize in a more or less fixed position beneath the rotor for this descent rate. Wind tunnel experiments would not generally permit such conditions. Two possibilities exist: Sufficient vorticity could continue to accumulate, increasing the self-propulsive speed of the ring, driving the ring clear of the rotor; or a disturbance could develop and remove the axially symmetric condition needed to sustain the ring's organization and ability to propel itself. Such distortions would lead to a mass-shedding of wake vorticity into the flow, destroying the ring formation altogether. In either case, the loss of the ring's influence allows a new ring to begin accumulating above the rotor and the process repeats.

Such an event is simulated in the ring emitter model by suddenly reducing the number of rings used in the calculation: At time = 87 s, the number of emitted rings is reduced to 80. However the ring remains beneath the rotor until the number of rings is reduced to 40 at time = 99 s. The remnant of the vortex ring now has insufficient self-propulsive speed and repositions itself above the rotor. Here the ring buildup process is reinitiated with up to 160 emitted rings permitted. The ring accumulates, drives itself through the rotor plane, and repeats the surging behavior seen in rotor thrust, even while the descent rate remains fixed. In this demonstration, it is clear that a wind tunnel mode experiment in VRS is very different from the free-flight rotor. Formation, dissipation, and reformation of a vortex ring in fixed descent rate wind tunnel testing produces sustained thrust fluctuations that do not represent the free-flight experience of Fig. 9.

The stability of the vortex ring above or below the rotor is worth commenting on. In the wind tunnel mode experiment, once the vortex ring transitions below the rotor, a significant reduction in the number of rings is needed to permit the accumulated ring to return to a position above the rotor. Similarly, a significant increase in accumulated rings is needed to allow the ring to pass from above the rotor plane to a position below the disk. The hysteresis is related to a similar effect observed in free descent. In the free descent of Fig. 9, it was observed that one final collective pitch reduction ($T = 550$ s) was needed to transition the vortex ring from a position below the rotor to a position above the rotor.

In a free descent, the position of the ring above the rotor is stable, as it is extremely difficult for this ring to return below the rotor by increasing collective pitch. The effect will be examined in the next section (where a climb in VRS is initiated); but for now, it is instructive to consider the behavior of newly emitted vortex rings and how they influence the stability of the accumulated ring's position.

As seen in the illustration of Fig. 13(a), if the balance of the accumulated vortex ring is "above" the rotor, all of the newly emitted rings will immediately expand in diameter due to the orientation of the accumulated vortex ring's circulation field. When an emitted vortex ring expands in diameter, its self-induced velocity is diminished by conservation of vorticity (recall the dual smoke ring analogy). Consequently, the emitted ring's ability to propel itself downward (against the surrounding flow) is reduced, causing the emitted ring to rise upward with the local flow and adding to the vorticity accumulation above the rotor. Conversely, in Fig. 13(b), when the accumulated vortex ring resides below the rotor, all newly emitted rings are forced to a smaller diameter. This increases their self-propulsive speed, allowing them to descend and reinforce the accumulation below the rotor. In this manner, the initial position of the accumulated vortex ring acts as a "switch" mechanism that strongly favors reinforcing the existing state on either side of the "negatively damped" region of Fig. 7. A converse effect is that of the newly emitted rings on the established vortex ring. While the accumulated ring is above the rotor, the circulation of all newly emitted rings will tend toward reducing the diameter of the accumulated ring. When sufficient rings pile on, the ring transits through the rotor plane when the accumulated ring's self-propulsive speed exceeds the free stream. The ring's arrested descent below the rotor plane is now caused because the newly emitted vortex rings (below the rotor) tend to increase the diameter of the established ring, so that its self-propulsive speed merely matches the free stream.

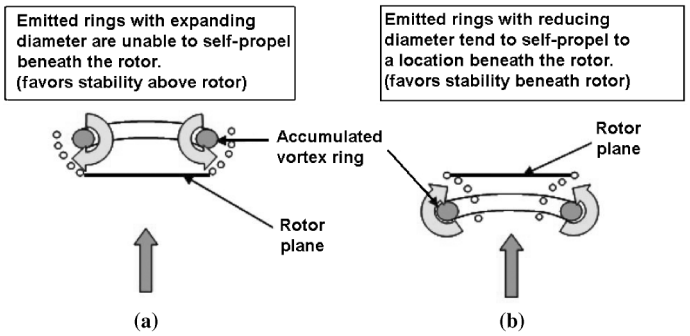


Fig. 13. The origin of hysteresis in stable ring positions: (a) Accumulated ring above the rotor causes newly emitted rings to "expand," whereas (b) accumulated ring below the rotor causes newly emitted rings to contract. The diameter of newly emitted rings is the main factor in determining the direction of subsequent ring convection and provides a mechanism that strongly favors keeping the current position of the accumulating ring.

Fighting VRS: Example of Increasing Collective Pitch

In a descent toward VRS, gradually adding collective pitch to the rotor does not stop the addition of tip vortices to the ring. Therefore, a straightforward "add more thrust" approach to tackling VRS is not likely to succeed. A higher collective pitch simply allows the rotor to equalize an even stronger vortex ring (and achieve higher descent rates) before transition occurs. To examine the "free-flight" behavior of increasing collective pitch during VRS, the ring emitter model is started in a windmilling condition and the collective pitch is increased incrementally, as seen in Fig. 14.

In Fig. 14, the free-flight rotor starts in a windmill state (at $T = 0$) with a descent speed that is roughly twice the hover-induced velocity. To arrest the descent rate, the collective pitch is gradually increased as shown in Fig. 14(a). As seen in Fig. 14(c), the thrust initially increases and then trends toward the $1g$ hover value. While the wake pattern remains columnar (and above the rotor), the descent speed decreases smoothly with collective pitch as one would expect in the positively damped regime. However, after $T = 33$ s (collective pitch $\sim$13 deg), further increases in collective pitch cease to arrest the descent rate. This point marks the formation of an accumulating ring vortex above the rotor disk. The growing induced flow of the vortex ring suppresses any increase in effective blade pitch angle (due to collective pitch increases). Any improvement in moderating the descent rate from the next 7 deg of collective pitch increase (from $T = 33$ s to $T = 85$ s) is completely canceled by the presence of the ring's inflow. In this time period, moderate thrust fluctuations ($\pm$20% of the mean thrust) are observed, yet the vortex ring remains above the rotor. At 20 deg of collective pitch, the rotor finally climbs through the vortex ring, ultimately achieving a vertical climb rate that well exceeds the hover inflow velocity.

The inability to arrest the descent rate over the majority of Fig. 14 is consistent with pilot reports of the collective pitch being ineffective during VRS. The only inconsistent result is that the simulation is eventually able to "climb out" of VRS. Note, however, that the collective pitch range available in the model may not represent normal helicopter

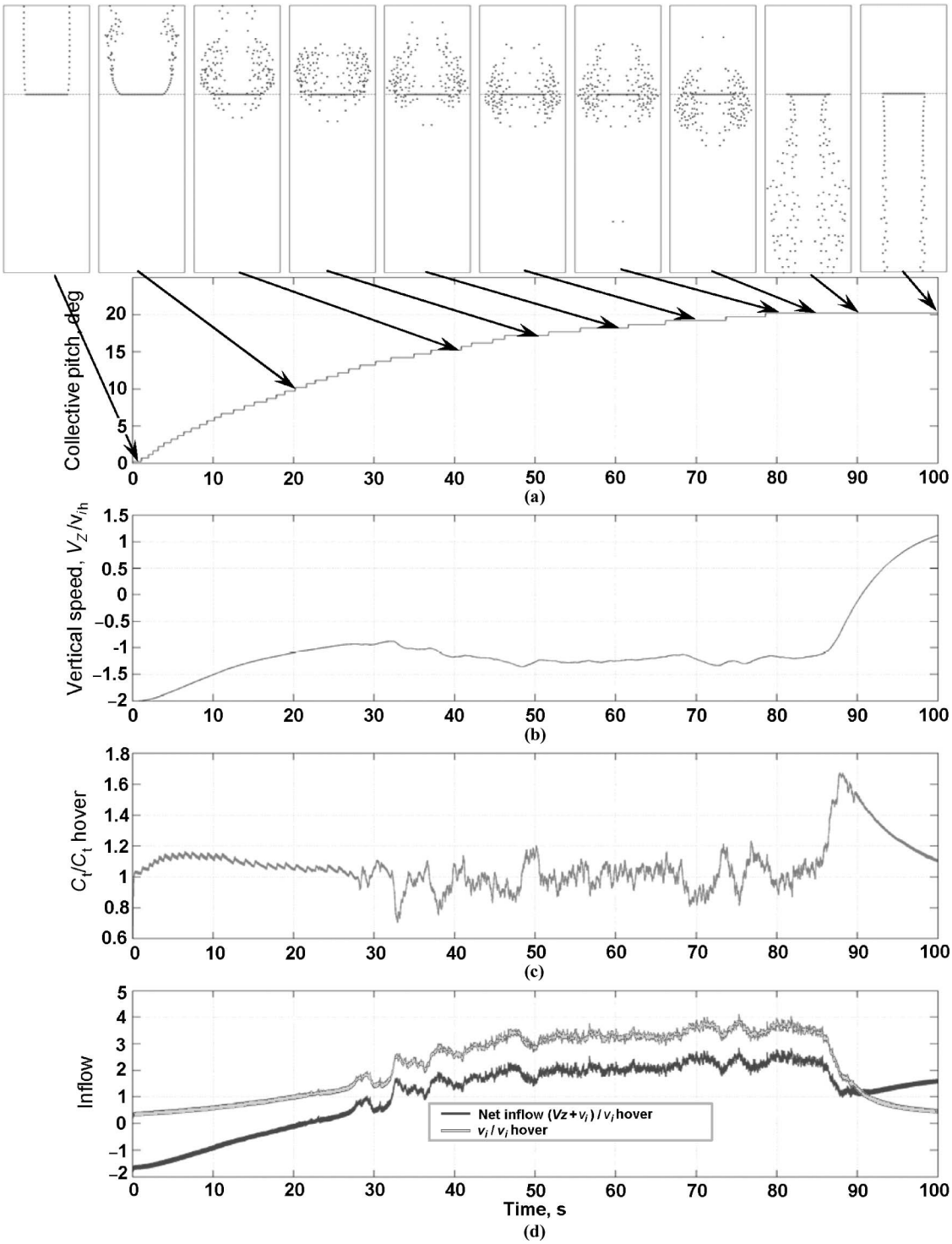


Fig. 14. Climbing out of VRS by increasing collective pitch may be possible if sufficient collective pitch range and power are available.

designs and therefore the result is unlikely to be achieved in conventional rotorcraft.

The ring emitter model is clearly able to predict the VRS initiation process and trace the negatively damped region to the organized accumulation of vorticity and its transit through the rotor plane. The question of whether it accurately predicts the outcome of a large (20-deg) collective pitch increase in a descent could be investigated with a suitable wind tunnel model. While the modeling assumptions are fairly general, it should be stated once more that the ultimate strength of the accumulated ring is mathematically limited by the number of rings used in the calculation. Nevertheless, on the basis that VRS (in general) does not extend to higher and higher descent rates, such a limitation appears necessary. A nonviscous mechanism for limiting the accumulated ring's ultimate strength was given by Ahlin and Brown (Ref. 10), but turbulence and the inability to maintain axial symmetry in the basic ring structure will also play an important role in limiting ring buildup. In either case, blade stall

is not the mechanism of thrust loss in VRS, and since the blades continue to produce thrust in proportion to their effective angle of attack, it seems reasonable to expect that a climb rate could be achieved—per the simulation—but only if power and collective pitch ranges are available.

Maneuvering in VRS

In all cases, the strategy for exiting VRS is to create distance between the established ring and the rotor. In Ref. 2, Kisor et al. reported that maneuvering in the V-22 suppressed the appearance of VRS. The suppression of VRS in accelerated flight has since been reported in the analytical work of Ahlin and Brown (Ref. 15). An explanation of this behavior is made here on the basis of considering the ring and rotor as separate entities. A vortex ring can propel itself only in a single axial direction. To participate in a maneuver, new rings (that are oriented to the new direction of flight) must be formed. The old established ring will continue on the original trajectory, separating itself (and its influence) from the rotor. This offers an insight into why the V-22 test team could penetrate deeply into the quasi-steady VRS boundary with no signs of VRS during dynamic maneuvers. If a rotor can displace itself from the ring or accelerate in a direction that the established ring cannot follow, the rotor will tend toward VRS-free conditions.

In part, this also explains why VRS cannot extend to higher forward speeds. In forward flight, the ring can stay with the rotor only if the ring's axis is effectively tilted so that its self-propulsion has a forward speed component. Alternatively, the leading edge of the ring must be replenished with vorticity as fast as the trailing edge is depleted. In a sustained "edgewise" acceleration, new tip vortices would have to continuously rebuild portions of the "old" established ring that did not follow the rotor. Clearly, it is much more difficult to intensify a vortex ring in an accelerated condition. The greater the acceleration (due to a turn), the deeper the rotor can penetrate into the boundary of descent rates and low speeds that would produce VRS in steady flight conditions. A sudden reorientation of the rotor tip path plane effectively establishes a new thrust vector and introduces an acceleration that does not support ring accumulation. Acceleration drives the rotor away from any ring formation that was formed in the quasi-steady preaccelerated descent. This is likely the reason why the action of nacelle tilt proved to be such a powerful mechanism for recovering from the VRS condition during V-22 flight trials.

The Future of VRS Research

Most wind tunnel tests of VRS have placed a model rotor at fixed collective pitch into a steady free stream. Steady-state wind/water tunnel tests show significant mean thrust variations with the descent rate and sustained thrust fluctuations as large as 80% of the mean thrust in steady descents. When the model rotor encounters VRS, thrust loss is measured, but the free-stream speed that is creating the descent rate remains unchanged and the rotor continues to descend at the same rate with a substantially reduced thrust. Flight test results and the ring emitter simulations show that this static behavior is not representative of an actual rotorcraft in VRS flight conditions. For a free-flying rotor, increased sink rate must accompany thrust loss, which completely changes the relationship between the free stream, the ring, and the rotor. Flight test results for the V22 (Ref. 2) showed that VRS is a relatively high-thrust phenomenon where the mean thrust rarely dropped below 90% of 1g normal load factor. Results also showed a smooth progression into fully developed VRS with none of the repeated thrust-surging behavior seen in wind tunnel tests.

Although wind tunnel testing with a free-stream speed that responds to thrust loss would be a challenge, such experimental results would provide a more accurate assessment of the growth, development, and behavior of VRS in free flight. A dynamic test rig may be able to facilitate such testing and could lead to a more rigorous experimental and theoretical basis for suppressing VRS. Results may offer added details on why maneuvers, control inputs, and the action of nacelle tilting (in the V-22) are such powerful mechanisms for extracting the aircraft from fully developed VRS. The power of flow visualization in examining rotor flow states should not be overlooked. Flow patterns can reveal an exceptional amount of detail concerning flow content. The flow visualization experiments of Drees and Hendal (Ref. 13) were completed more than 55 years ago, and this work has been cited in almost every publication concerning VRS research. It would be a missed opportunity to omit flow visualization from a new experimental test program.

Conclusions

1) The fundamental dynamic of VRS is the propensity for vortex wake elements to accumulate in a highly organized structure that multiplies the induced flow experienced by the descending rotor. Rotor inflow increases with VRS, causing the blades to operate at a reduced effective angle of attack and lose thrust. While wake filament disorganization can occur as the wake ages, it is not the characterizing feature that enables VRS. The organized part of the wake is responsible for increasing the rotor inflow, and inflow will reduce the effective blade angle of attack. The organized accumulation of the wake (in a low forward air speed descent) is therefore the characterizing feature of VRS.

2) Steady speed wind tunnel tests do not appropriately simulate the primary, unsteady event of VRS (where the ring transitions through the rotor plane a single time). Steady conditions can exist prior to, and after, transition—if the accumulated vortex ring is maintaining its position relative to the rotor. However, even in these steady conditions, wind tunnel results will typically show that the thrust is below that required for sustained 1g flight. In free flight, thrust loss is mitigated by higher descent rates that are precluded by a fixed wind tunnel speed. For a rotorcraft in a free descent, VRS remains a relatively high thrust condition until the ring transitions through the rotor plane.

3) The transition of the (vortex) ring from a position below the rotor to a position above the rotor is the "main event" in VRS. It is associated with a thrust surge that happens once in a free descent but repeats continuously in "steady speed" wind tunnel testing. The unsteadiness perceived in wind tunnel tests comes primarily from a stable structure (the vortex ring) undergoing repeated relative motion through the rotor disk. The simple ring emitter model clearly predicts the VRS initiation process and traces the origin of the negatively (thrust) damped region to an accumulating vortex ring that undergoes relative motion through the rotor. The ring's accumulation and its eventual transit through the rotor disk causes the blades to operate at progressively lower effective angles of attack and lose thrust.

4) The stable relationship between thrust and descent rate/inflow on either side of the negatively damped region (of Fig. 7) has been related to stable wake formations where organized vorticity is added to the vortex ring as quickly as it is depleted/dispersed/disorganized. When the toroidal wake is not accumulating organized vorticity, the descent rate relationship with rotor thrust is positively damped.

5) VRS is a perturbation from the upper (working-state) branch of simple momentum theory's inflow state curves. The transition of the vortex ring through the rotor is not associated with the transition to the windmilling branch.

6) The accumulated vortex ring takes time to develop and once established can only propel itself in a single axial direction. The rotor can maneuver, but the ring cannot. Therefore, maneuvers suppress the effects of VRS.

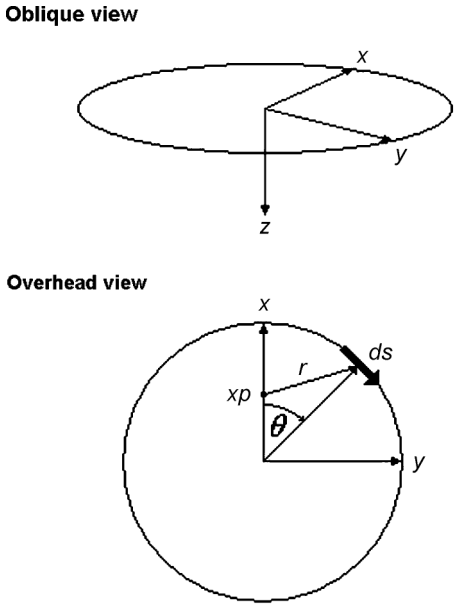


Fig. A.1. Geometry of a vortex ring.

Appendix

The ring emitter model uses the differential form of the Biot–Savart law to find the induced velocity at an arbitrary point near a vortex ring. Consider Fig. A.1, which shows a ring in the x–y plane centered at the origin of a reference axis system. The positive direction of a differential element of the vortex filament dS points in the clockwise direction. Let the parameter θ measure the distance around the ring, starting at the x-axis and proceeding positively around the z-axis.

If R is the distance from the origin to the vortex ring, then the direction of the differential segment of the vortex ring is given by

$$dS = \left\{ \begin{array}{c} -R\sin\theta \\ R\cos\theta \\ 0 \end{array} \right\} d\theta \tag{A.1}$$

A presumption of axisymmetric flow means that the point of interest need only move along the x-axis and z-axis to fully describe the flow anywhere near the ring. The vector from "xp" to the differential segment of the ring is

$$r = \left\{ \begin{array}{c} R\cos\theta - x_p \\ R\sin\theta \\ -z_p \end{array} \right\} \tag{A.2}$$

The circulation strength of the vortex, when emitted from the rotor disk, is γ and is a constant. Therefore, the differential induced velocity is

$$dv_i = \frac{\gamma}{4\pi}\frac{r \times dS}{\|r\|^{3/2}} \tag{A.3}$$

Define the parameter$L^2 = R^2 + x_p^2 + z_p^2$. Then the induced velocity at point "p" is

$$\begin{aligned} Vi_x &= \frac{\gamma}{4\pi}\int_0^{2\pi} \frac{R\cdot z_p\cdot\cos\theta\, d\theta}{(L^2 - 2R\cdot x_p\cdot\cos\theta)^{3/2}} \\ Vi_z &= \frac{\gamma}{4\pi}\int_0^{2\pi} \frac{(R^2 - R\cdot x_p\cdot\cos\theta)\, d\theta}{(L^2 - 2R\cdot x_p\cdot\cos\theta)^{3/2}} \end{aligned} \tag{A.4}$$

Employing the following definitions:

$$\begin{aligned} A_x &= \frac{R\cdot x_p}{L^3} \\ A_z &= \frac{R\cdot z_p}{L^3} \\ B &= \frac{2R\cdot x_p}{L^2} \\ D &= \frac{R^2}{L^3} \end{aligned} \tag{A.5}$$

the induced velocity is expressed in the concise form

$$\begin{aligned} Vi_x &= \frac{\gamma}{4\pi}(A_z I_C) \\ Vi_z &= \frac{\gamma}{4\pi}(DI_1 - A_x I_C) \end{aligned} \tag{A.6}$$

where

$$I_1 = \int_0^{2x} \frac{d\theta}{(1 - B\cos\theta)^{3/2}} \tag{A.7}$$

$$I_C = \int_0^{2\pi} \frac{\cos\theta\cdot d\theta}{(1 - B\cos\theta)^{3/2}} \tag{A.8}$$

The x-direction induced velocity will act on neighboring rings, causing them to shrink or expand radially. The z-direction induced velocity acts on neighboring rings, pushing them away or attracting them according to the sign. The z-direction induced velocity also acts on the generating ring and is responsible for the self-propel velocity normal to the plane of the ring.

The solutions to I_1 and I_C involve elliptical integrals of the first and second kind but are solved numerically in this investigation using B as a parameter. Both integrals are undefined when $B = 1$, which corresponds to point "p" lying on the ring. Scully, as reported by Wachspress et al. (Ref. 16), modifies the two-dimensional Biot–Savart law to account for a rotational core, which permits evaluation on the ring itself.

$$v_i = \frac{\gamma}{2\pi}\frac{r}{\left(r^2 + r_{\text{core}}^2\right)} \tag{A.9}$$

Extending Scully's modification to three dimensions, the modified differential form of the Biot–Savart reads this way:

$$dv_i = \frac{\gamma}{4\pi}\frac{r \times dS}{\left(|r|^2 + r_{\text{core}}^2\right)^{3/2}} \tag{A.10}$$

The solution changes only the definition of the parameter L:

$$L^2 = R^2 + x_p^2 + z_p^2 + r_{\text{core}}^2 \tag{A.11}$$

The integrals I_1 and I_C were evaluated as a function of core size when the point of interest was situated on the ring. The vertical velocity thus obtained is the self-propel velocity of the ring. The results compare well with the analytical solution by Saffman (Ref. 11).

$$V_z = \frac{\gamma}{4\pi R}\left[\ln\frac{8R}{r_c} - \frac{1}{4} + o\left(\frac{r_c}{R}\right)\right] \tag{A.12}$$

The factor $o(r_c/R)$ in Saffman accounts for variation from circular cross-section. This model is perfectly circular, so that factor is nil. The factor of 1/4 accounts for a core which is uniform or rotational. This factor increases to 1/2 if the core is hollow.

The rotor has n_b blades and generates thrust, T. This means that each blade produces lift T/n_b. Each blade sheds a vortex ring at blade passage

frequency. The Kutta–Joukowski law estimates the strength of the vortex.

$$\frac{T}{n_b R} = \rho V_{\text{tot}} \Gamma_0 \tag{A.13}$$

This model uses the total velocity at the tip of the rotor, i.e.,

$$V_{\text{tot}} = \left[V_{\text{tip}}^2 + (w_0 - w_i)^2 \right]^{1/2} \tag{A.14}$$

where $V_{\text{tip}} = \Omega R$, w_0 is the vertical inertial velocity of the rotor, and w_i is the vertical induced velocity.

The core size is specified by the user.

When the diameter of a ring changes due to a radial component of induced velocity, the vortex is stretched. Since circulation strength is a constant, the vorticity must vary as diameter varies. Consider the ring as a torus. Conserving angular momentum is tantamount to preserving the volume of the torus. Thus, the core size varies in the following way as torus radius changes:

$$r_{\text{core}} = r_{\text{core}\,0} \sqrt{\frac{R_0}{R}} \tag{A.15}$$

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